Problem Set 8

Due Saturday, November 7, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets. Use of any AI or LLM tools is strictly forbidden.

1. Consider a hydrogenic atom (or ion) with nuclear charge Ze and a single atomic electron.

   (a) Setting any constant parameters to one and $r = 1$ and using Mathematica's SphericalPlot3D command, make 3D plots for the angular parts of $|\psi_{200}(r,\theta,\phi)|^2$, $|\psi_{210}(r,\theta,\phi)|^2$, $|\psi_{211}(r,\theta,\phi)|^2$, and $|\psi_{21-1}(r,\theta,\phi)|^2$. Which ones have spherical symmetry (independent of θ and ϕ)? Which ones have azimuthal symmetry (independent of ϕ)? In which states is the electron not likely to be found (i). along the $z-$axis, (ii). on the $x-y$ plane? (5 points)

   (b) Evaluate $\langle r\rangle$ and $\langle r^2\rangle$ for the ground state. (5 points)

   (c) Evaluate $\langle x\rangle$ and $\langle x^2\rangle$ for the ground state. (5 points)

   (d) Determine the most probable value of r in the ground state, and write its numerical value. How would you interpret what this number means? (5 points)

2. (a) Show that all s state (or $l = 0$) energy eigenfunctions of hydrogenic atoms or ions have a maximum at $r = 0$. (5 points) [As we have seen in problem 1 above, quantities for hydrogenic atoms or ions follow from those for hydrogen atom with $e^2 \to Ze^2$.]

   (b) Show that all non-s state (or $l \neq 0$) energy eigenfunctions of hydrogenic atoms or ions vanish at $r = 0$. (5 points)

   (c) What is the most probable radius for a "circular state" of a hydrogenic atom or ion? (5 points) [A circular state has the maximum value of l and m_l for a given n.]

   (d) What does the probability distribution look like for a circular state? (5 points)

3. Determine the matrices for $\hat{\vec{L}}^2$, $\hat{L}_z$, $\hat{L}_x$, and $\hat{L}_y$ for the four $n = 2$ states of hydrogen in the bases $|nlm_l\rangle = |200\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$, $|211\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}$, $|210\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}$, and $|21-1\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$ and the commutation relation among them. (5 points)

4. Consider a state made from direct products of angular momenta with $j_1 = 1$ and $j_2 = \frac{3}{2}$: $\sqrt{\frac{3}{5}}\,|1,-1\rangle \otimes |\frac{3}{2},-\frac{1}{2}\rangle + \sqrt{\frac{2}{5}}\,|1,0\rangle \otimes |\frac{3}{2},-\frac{3}{2}\rangle$.

   (a) Show that the given state is a simultaneous eigenstate of $\hat{\vec{J}}^2$ and $\hat{J}_z$, and write down the numerical values of the corresponding eigenvalues. (5 points)

   (b) Applying the raising/lowering operators, construct a normalized state which has an eigenvalue for $\hat{J}_z$ that is bigger by one unit of $\hbar$. (5 points)

   (c) Applying the raising/lowering operators, construct a normalized state which has an eigenvalue for $\hat{J}_z$ that is smaller by one unit of $\hbar$. (5 points)

5. How long did it take you to complete this problem set?